Edward Ajayi

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Education

Carnegie Mellon University
Master of Science in Engineering Artificial Intelligence — *Aug. 2024 – Present*

- **Relevant Courses:** Natural Language Processing, Deep Learning, Question Answering with LLMs
- **Specializations:** Data Science, Machine Learning, NLP

Ekiti State University
Bachelor of Science in Computer Science — *Oct. 2014 – Nov. 2019*

- **Specializations:** Computer Science

Research Interests

Socially Aware AI, Multimodal Reasoning, Computational Humor, Culturally Grounded NLP, AI for Healthcare

Selected Publications

1. **E. Ajayi**, M. Kachweka, M. Deku, and E. Aiken. *A Machine Learning Approach for Detection of Mental Health Conditions and Cyberbullying from Social Media.* In *Proceedings of Machine Learning Research (PMLR)*. Accepted for Oral Presentation at the *AAAI-26 Bridge Program on AI for Medicine and Healthcare (AIMedHealth)*. [Paper Link]
2. **E. Ajayi** and P. Mitra. *Automatic Humor Detection: From Theoretical Foundations to Large Language Models.* To be submitted to *ACM Computing Surveys*. [Manuscript Link]
3. **E. Ajayi**, B. Tadele, E. Umwari, M. Deku, P. Singadi, C. Edeh, and J. Udahemuka. *Language Diversity: Evaluating AI Performance on African Languages.*

Experience

Research Assistant
Spatial and Language Technologies Lab, CMU-Africa — *Aug. 2025 – Dec. 2025*

- Conducting research on NLP for automatic humour detection and generation, dataset preparation, model development, and evaluation.
- Contributing to projects on low-resource language technologies and advancing cultural understanding in large language models.

Research Intern - Business Automation
CMU-Africa — *Jun. 2025 – Aug. 2025*

- Developing and prototyping an agentic AI framework to automate complex business workflows.
- Researching and implementing AI safety protocols and robust LLM reasoning for ethical alignment and deployment.
- Analyzing agent performance to establish benchmarks for human-in-the-loop oversight and develop explainability metrics to support transparent corporate accountability.

Client Service Officer
Stanbic IBTC Bank — *Dec. 2021 – Jul. 2024*

- Leveraged data-driven insights to cross-sell over N400M in assets, enhancing client trust and digital adoption by 25%.
- Managed client relationships, identifying service gaps to improve operational efficiency.

Operations Analyst
PricePointe — *Sep. 2021 – Dec. 2021*

- Analyzed pricing strategies with SQL and Python, boosting revenue by 20% while maintaining customer loyalty.

Technical Skills

Programming & Scripting: Python, SQL, Bash
Data Science & ML: Pandas, NumPy, Scikit-Learn, PyTorch, Matplotlib, Seaborn, Hugging Face Transformers
Natural Language Processing: Langraph, OpenAI Agents SDK, NLTK, spaCy, Sentiment Analysis, LLM Fine-Tuning, Prompt Engineering, Retrieval-Augmented Generation (RAG)
Data Engineering & Tools: Jupyter, Neo4j, FastAPI, BeautifulSoup, Google Sheets API, MS Excel
Cloud & Deployment: AWS, Docker, REST APIs

Projects

AI Workflow Automation Agent | *Python, LangChain, FastAPI, Google Sheets API, SMTP* *Jul. 2025*

- Built an autonomous agent that interacts with a Google Form–linked Sheet to process business requests.
- Sends email notifications to approvers, who review AI-generated content via a custom dashboard.
- Integrated a QA bot to monitor and update sheets based on approvals and feedback.

Multichannel HR Assistant Bot | *RAG, Python, Slack API, FastAPI, LangChain* *Aug, 2025*

- Developing a Retrieval-Augmented Generation HR assistant accessible via Slack, email, and web.
- Supports employee queries using secure internal document retrieval and conversational response generation.

Kigali Smart City Guide | *Python, Neo4j, LangChain, Llama* *Nov. 2024*

- Built a Retrieval-Augmented Generation chatbot, ensuring safe and context-aware responses for urban use cases.

Soft Skills

Critical Thinking, Problem-Solving, Collaboration, Communication, Adaptability

Training & Conferences

Participant, Deep Learning Indaba 2025 Rwanda
Deep Learning Indaba *Aug. 2025*

- Attended workshops on deep learning, AI for social good, and research collaboration across African universities.

Participant, Digital Humanities Summer School Rwanda
DIGIHUM/CMU-Africa *Jul. 2025*

- Engaged in computational methods for humanities research, including NLP applications.

Participant, ML Summer School Online
Cohere *Jul. 2025*

- Intensive sessions on machine learning, foundation models, multimodal learning, and AI agents.

Learner, Natural Language Processing Specialization Online
Coursera, DeepLearning.AI *May 2025*

- Completed 4-course specialization covering NLP, sequence models, attention, and transformers.

Volunteering/Leadership

President, CMU-Africa Research Club Kigali, Rwanda
Carnegie Mellon University Africa *Jan. 2025 – Dec. 2025*

- Led 12 major events in 2025 supporting student research.
- Launched website with research guides, conference updates, and collaboration hub.
- Organized first Student Research Showcase with 20+ posters and workshops.
- Hosted three-part PhD Application Series, faculty sessions, and alumni panels supporting 40+ students.
- Collaborated with ACM SIGCOMM and Upanzi Network for remote conference participation.

Peer Reviewer, AAAI AIMedHealth Bridge 2026 Remote
AAAI Symposium on AI for Medicine and Health *Nov. 2025 – Nov 2025*

- Reviewed submissions in applied machine learning and AI for healthcare under the AAAI AIMedHealth Bridge 2026 program.

IEEE Student Member CMU-Africa
Institute of Electrical and Electronics Engineers *Feb. 2025 – Present*

- Collaborating with IEEE members to advance research initiatives among club members.

Certifications

- Microsoft Certified Azure Data Scientist Associate - View
- Microsoft Certified Azure Data Fundamentals - View
- DataCamp Python Data Associate - View

Honors and Awards

- Mastercard Foundation Scholarship, Carnegie Mellon University Africa
- Federal Scholarship Board Award for Undergraduates, Ekiti State University